

Problem

A loudspeaker is connected to an amplifier as shown in Fig. 2.139. If a $10\text{-}\Omega$ loudspeaker draws the maximum power of 12 W from the amplifier, determine the maximum power a $4\text{-}\Omega$ loudspeaker will draw.

Figure 2.139:



Step-by-step solution

Step 1 of 2

Refer to the circuit in Figure 2.139 in the textbook.

The expression for the maximum power drawn by the loud speaker is,

$$P = \frac{V_{rms}^2}{R}$$

$$\text{Since } V_{rms} = \frac{V_m}{\sqrt{2}},$$

$$\begin{aligned} P &= \frac{\left(\frac{V_m}{\sqrt{2}}\right)^2}{R} \\ &= \frac{V_m^2}{4R} \end{aligned}$$

Substitute 12 W for P and $10\text{ }\Omega$ for R .

$$12 = \frac{V_m^2}{4 \times 10}$$

$$V_m^2 = 12 \times 40$$

$$V_m = 4\sqrt{30}\text{ V}$$

Step 2 of 2

The maximum power drawn by the loud speaker is,

$$P = \frac{V_{rms}^2}{R}$$

Since $V_{rms} = \frac{V_m}{\sqrt{2}}$,

$$P = \frac{V_m^2}{4R}$$

Substitute $4\sqrt{30}$ V for V_m and $4\ \Omega$ for R .

$$\begin{aligned} P &= \frac{(4\sqrt{30})^2}{4 \times 4} \\ &= \frac{16 \times 30}{16} \\ &= 30\text{ V} \end{aligned}$$

Thus, the maximum power drawn by $4\ \Omega$ loud speaker is 30 V.